Electrochemical reactions of white phosphorus*

D. G. Yakhvarov,[a]* E. V. Gorbachuk,[b] R. M. Kagirov,[a] and O. G. Sinyashin[a]

[a]A. E. Arbuzov Institute of Organic and Physical Chemistry, Kazan Scientific Center, Russian Academy of Sciences, 8 ul. Arbuzova, 420088 Kazan, Russian Federation.
Fax: +7 (843) 273 2253. E-mail: yakhvar@iopc.ru
[b]Kazan (Volga Region) Federal University, 18 ul. Kremlevskaya, 420008 Kazan, Russian Federation

The main electrochemical transformations of elemental (white) phosphorus were considered. Special attention was given to the recently developed processes of preparation of organophosphorus compounds (OPCs) with phosphorus—carbon bonds. The electrochemical approaches to the synthesis of OPCs from white phosphorus using organonickel and organozinc reagents are described. The importance of using the electrochemical methods for the generation of highly reactive phosphorus intermediates was shown for phosphine oxide H_3PO obtained for the first time. This provides significant prospects for the electrochemical approaches that could be applied for the development of technologies of the chlorine-free synthesis of OPCs from white phosphorus.

Key words: white phosphorus, activation, electrosynthesis, electrochemistry, phosphine oxide, phosphorus compounds.

Development of novel technologies for synthesis of phosphorus-containing compounds (PCCs), including organophosphorus compounds (OPCs), is a key task of the modern chemistry of phosphorus.[1,2] Elemental (white) phosphorus is the main starting reagent for the preparation of various OPCs widely used in chemical and food industries, biology, and medicine.[2—4] The modern methods for OPCs synthesis are based on "chlorine" technologies that include the use of phosphorus acid chlorides formed upon the chlorination of white phosphorus by gaseous chlorine. This technology is energy consuming and ecologically dangerous because of the evolution of toxic substances (HCl, $POCl_3$, PCl_5, and others) during the process.

The reactions using transition metal complexes[5,6] and electrochemical methods[7] are alternative approaches to the preparation of organophosphorus compounds. A combination of the methods of metal complex catalysis[8,9] and organic electrosynthesis[10] attracts increasing attention due to high selectivity and efficiency of the considered approach for preparation of diverse chemical compounds with carbon—carbon and element—carbon bonds. Mild conditions of the process, one-stage character, and cyclic recovery of the catalyst, as well as the use of a convenient and relatively cheap type of energy, *viz.*, electricity, are advantages of electrocatalytic methods. The application of electrochemistry in preparative synthesis provides considerable prospects in the development of future chemical technologies. This is due to the fact that the electrochemical approach makes it possible to selectively generate various highly reactive intermediates, to enhance or diminish the activity of the substrate in reactions with electrophiles or nucleophiles when using an electron capable of *in situ* reducing (cathodic process) and oxidizing (anodic process) the substrate at the corresponding electrode potential. In addition, the method considered allows one to control the process and *in situ* detect short-lived intermediates, to study their electrochemical properties, and to determine the kinetics and mechanism of the process, which is one of the most important tasks of homogeneous metal complex catalysis.

The use of transition metal complexes[11—13] in the activation and selective transformation of a white phosphorus molecule into organophosphorus compounds is an efficient alternative for the current methods of industrial synthesis of OPCs. The application of electrochemical methods for the selective generation of highly reactive organometallic intermediates capable of reacting with white phosphorus and organic substrates would make it possible to perform chemical transformations under "green" conditions with the entire control of the process due to the variation of the working electrode potential and current density.[7]

Various electrocatalytic processes involving transition metal complexes have been developed to present.[14,15] It was found that highly reactive organometallic σ-complexes

* Based on the Materials of the International Congress on Organic Chemistry dedicated to the 150th anniversary of the Butlerov´s Theoty of the Chemical Structure of Organic Compounds (September 19—23, 2011, Kazan, Russia).

Published in Russian in *Izvestiya Akademii Nauk. Seriya Khimicheskaya,* No. 7, pp. 1285—1298, July, 2012.

containing metal—carbon bonds are the key intermediates in homo- and cross-coupling processes involving organic halides, chlorophosphines, and some other organic and organoelement compounds.[16]

At the moment scientific world literature includes restricted data on the properties, structures, and reactivity of organonickel σ-complexes, the mechanisms of activation and transformation of white phosphorus tetrahedron in the coordination sphere of the transition metal, and phosphorus intermediates formed during electrochemical processes involving white phosphorus. The σ-complexes are considered[17—19] only as intermediates of various catalytic processes involving the nickel complexes. Studies devoted to the selective preparation of organometallic σ-complexes and complexes containing the transformed molecule of white phosphorus and to investigation of their properties and reactivity are almost absent, while there is few data on using electrochemical methods in these processes. The study of the reactivity of the organonickel σ-complexes can serve as a good theoretical basis in understanding the mechanisms of many catalytic processes involving nickel complexes. The possibility of using the electrochemically synthesized organonickel σ-complexes for the solution of basic problems of the modern organoelement chemistry related to the development of new methods for synthesis of organophosphorus compounds is of special interest.

In this review we generalized the currently available data on the electrochemical processes involving elemental (white) phosphorus. The main attention is concentrated on the description of electrochemical reactions involving elemental (white) phosphorus and the study of the structures and reactivity of intermediates of the P_4 transformation into organophosphorus compounds with the P—C bonds.

1. Electrochemical reactions of elemental (white) phosphorus

The study of the electrochemical behavior of white phosphorus and the possibilities of electrosynthesis based on white phosphorus is at present rather poorly developed direction, although these studies have been started more than fifty years ago.[20] However, the number of works in this area is small.[5,7] The systematic study of the reactivity of the white phosphorus molecule is necessary to reveal routes of its reactions, which would provide the target choice of conditions for electrochemical synthesis of OPCs from elemental (white) phosphorus.

1.1. Electrochemical reduction

The first works in the field of electrochemistry of elemental (white) phosphorus were made in Russia in the research group guided by Prof. A. P. Tomilov (Moscow) (see Ref. 21) and were later continued in the research group of Prof. Yu. M. Kargin in Kazan (see Refs 22 and 23). Taking into account wide propagation of polarography that studies electrochemical processes at a mercury electrode, the first electrochemical experiments with P_4 were carried out under these conditions. It was established that a white phosphorus molecule accepts an electron at the potential $E_{1/2} = -1.55$ V (*vs* bottom mercury in DMF). This electrochemical process is irreversible and results in the formation of a radical anion (Scheme 1) capable of further chemical transformations.[21]

Scheme 1

$$P_4 + e^- \longrightarrow [P_4]^{\cdot -}$$

It is shown in the recent work of Yu. H. Budnikova and M. K. Kadirov that the $P_4{}^{\cdot -}$ radical anion can be detected by the ESR method as a spin adduct with α-phenyl-*N*-*tert*-butylnitrone.[24] This experiment was carried out in a special electrochemical ESR cell with the spiral platinum working electrode in the potentiostatic mode. The electrochemical behavior of white phosphorus in alcohol solutions was described in detail.[25]

The phosphide anions formed by electrochemical reduction can initiate the polymerization of white phosphorus forming polyphosphorus compounds insoluble in organic solvents. However, in the presence of active proton donors capable of protonating the phosphide anions initially originated in the electrochemical process, the derivatives with P—H bonds are formed. During this process, successive opening of all P—P bonds of the white phosphorus tetrahedron occurs to form phosphine PH_3 as the major electrolysis product[26] (Scheme 2).

Scheme 2

$$P_4 \xrightarrow{12\ H^+,\ 12\ e^-} 4\ PH_3$$

It was found that the nature of the solvent and the material of the used electrode has a significant impact on the electrochemical reduction of white phosphorus. The review[27] gives the most complete representation of the influence of various factors on the preparation of phosphine from white phosphorus. It was shown[28] that white phosphorus is reduced through the intermediate chemisorbed product $P_4(Hg)_x$ if DMF was replaced by ethanol as the solvent (Scheme 3).

Scheme 3

$$P_4(Hg)_x \xrightarrow{12\ H^+,\ 12\ e^-} 4\ PH_3 + x\ Hg$$

The use of electrodes of metals with a high overvoltage of hydrogen reduction (Hg, Pb) also allows one to carry out the electrochemical synthesis of phosphine from white phosphorus in good yield.[29] The process was found to proceed through a series of consecutive stages of formation of highly reactive phosphorus hydrides of the $(PH)_x$ type. One of the directions of OPCs electrosynthesis from white phosphorus related to its reduction in a proton-donor medium was developed on the basis of these concepts.

Since the electrochemical reduction of elemental (white) phosphorus during the polymerization of P_4 affords polyphosphorus hydrides capable of further electrochemical reducing to phosphine, it was interesting to study the electrochemical properties of red phosphorus that is a polyphosphorus compound. It was established that the electrochemical reduction of red phosphorus at the lead cathode in an aqueous alkaline solution (NaOH, 15—20%) also results in the formation of phosphine,[30] the 60—83% yields (based on phosphorus) of the target product can be achieved. It should be mentioned that this reaction occurs in the temperature range from 70 to 100 °C. The change in the alkali concentration exerts a substantial effect on the course of electrochemical synthesis of phosphine. An increase in the alkali concentration in a solution results in the formation of anions of phosphorous (H_3PO_3) and hypophosphorous (H_3PO_2) acids in which the oxidation state of phosphorus is +3 and +1, respectively. The oxidized phosphorus derivatives are probably formed due to the disproportionation of white phosphorus in strongly alkaline solutions or by the electrochemical oxidation of phosphine, which, however, was not observed.

It should be mentioned that phosphorous acid H_3PO_3 is also formed in the electrochemical reduction of white phosphorus in aqueous solutions of hydrohalic acids.[31] The formation of phosphorous acid in this process can be explained by the electrochemical oxidation of chloride anions to molecular chlorine acting as a chlorinating agent with respect to a white phosphorus molecule. The subsequent hydrolysis of the P—Cl bond leads to the formation of compounds with P—OH bonds, and the rearrangement of $P(OH)_3$ to the tautomeric form of tetracoordinated phosphorus gives $HP(O)(OH)_2$ (phosphorous acid). It should be noted that hypophosphorous H_3PO_2 and phosphorous H_3PO_3 acids can also be obtained by the electrochemical reduction of white phosphorus in the presence of organic bases (ethylenediamine and triethylamine).[32]

1.2. Electrochemical oxidation

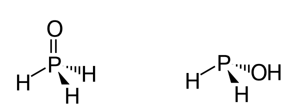
Only one work is devoted to the electrochemical oxidation of white phosphorus.[33] It was shown that phosphorus deposited on porous conducting matrices in neutral and acidic media is oxidized to phosphoric and phosphorous acids with the overall current efficiency about 100%. Probably, the process proceeds through intermediate compounds of an undetermined composition or is accompanied by the oxidation of phosphine formed due to the disproportionation of the P_4 molecule.

The indirect electrochemical oxidation of phosphorus was performed in alcohol solutions in an undivided electrochemical cell. The reaction of a phosphorus molecule with anodically generated chlorine affords intermediate chloride derivatives. The reactions of the obtained intermediates with water or alcohol give phosphorus-containing acids or trialkyl phosphates in 40—70% yields.[34]

In spite of noticeable progress in the chemistry of elemental phosphorus observed in recent years, researchers still search for the more complete involvement of phosphorus in diverse reactions and the enlargement of the scope of possible processes for the production of useful products. A necessary condition is the deep study of the mechanism of reactions of white phosphorus and the routes of further transformations of the primarily formed products in various reaction systems.

1.3. Electrochemical generation of phosphine oxide H_3PO

Although phosphorus oxides, oxyacids, and their esters are important compounds, which find wide application in the modern chemical industry and agriculture, participate in energy conversion in most organisms, and are used as additives to polymer materials and in drug preparation,[2] syntheses of these compounds are not a subject of intensive studies. These derivatives can easily be obtained by the chemical oxidation of white phosphorus with air oxygen (or other accessible chemical oxidants) and subsequent interaction with the organic substrate. However, it should be mentioned that some oxygen-containing phosphorus compounds can also be intermediates in the synthesis of OPCs with P—C bonds. For example, the oxygen-containing compounds with reaction centers P—H (phosphorous and hypophosphorous acids) can add to unsaturated bonds of organic substrates[35—39] giving organophosphorus products with P—C bonds. Unfortunately, poor attention was given until recently to species with a lowered oxidation state, such as PO, HPO, and P_2O, for which processes of synthetic isolation and direct proof of existence are rare.[40,41] In this respect, phosphine oxide H_3PO (**1**) is one of the species that is most difficult to observe.



1 **1´**

The formation of this molecule has been detected in the reaction of atomic oxygen with PH_3 using a discharge-flow system equipped with molecular-beam sampling mass spectrometry.[42] In other earlier described case, red-light photolysis of co-deposited PH_3/O_3 in the argon matrix at 12—18 K was used. Thus, the authors[43,44] succeeded to generate and capture **1** in the very low concentration along with its tautomer (phosphonous acid $H_2P(OH)$ (**1´**)) and identified it by FT-IR spectroscopy. The formation of H_3PO was also observed in argon matrices after the photochemical irradiation of a $VOCl_3$, CrO_2Cl_2, and PH_3 mixture.[45]

We found that this unstable (under normal conditions) compound, H_3PO (**1**), can easily be generated in solutions from white phosphorus using electrochemical methods.[46] Moreover, the experimental proofs for its stability in solution have first been obtained by ^{31}P NMR spectroscopy. However, this phosphorus compound could be isolated only as a ligand in the coordination sphere of the water-soluble ruthenium complexes after its tautomerization to **1´**.

The electrochemical generation of H_3PO was carried out in an undivided electrochemical cell equipped with a lead cathode and a soluble zinc anode using an emulsion of white phosphorus in a slightly acidic aqueous ethanol solution (water—ethanol, 2 : 1 vol/vol, with an additive of 2 *M* HCl) at 60 °C. The use of this system additionally confirms necessity of the complex approach to investigation of electrochemical reactions of white phosphorus, because the experimental conditions (solvent, material of the electrode, and temperature) exert a substantial effect on the mechanism of electrochemical reactions involving white phosphorus.

The general electrochemical process can be divided into two stages. The electrochemical generation of PH_3 on the lead electrode occurs at the first stage as described earlier,[29] whereas the second stage is the mild anodic oxidation of PH_3 to H_3PO on the zinc anode surface. The electrochemical hydrogenation of elemental (white) phosphorus to phosphine PH_3 has previously been studied in detail. In the optimized processes that occur on the lead cathode at 70 °C in basic media (≤15% NaOH) the yield of phosphine attains 80%[29,47]; however, the formation of phosphine oxide H_3PO was not detected.

The results of cyclic voltammetry (CV) that present the electrochemical properties of phosphine PH_3 generated in the cathodic process and giving the peak of irreversible electrochemical oxidation ($E_p^{ox} = +1.24$ V *vs* $Ag/AgNO_3$, 0.01 mol L^{-1} in MeCN) show that PH_3 is electrochemically active in the anodic potential range from +0.80 to +1.25 V (*vs* $Ag/AgNO_3$, 0.01 mol L^{-1} in MeCN) and can be oxidized in an acidic water—ethanol solution to H_3PO (Fig. 1).

The general electrochemical process resulting in the cathodic reduction of P_4 to PH_3 and anodic oxidation of PH_3 to H_3PO ($E_p^{ox} = +1.24$ V *vs* $Ag/AgNO_3$, 0.01 mol L^{-1} in MeCN) is shown in Fig. 2.

The effect of different operating conditions was studied to optimize the electrochemical process of H_3PO formation. The best results were obtained when solid P_4 (70 mg) was suspended in a water—ethanol mixture (2 : 1, 30 mL) in an electrochemical cell heated to 60 °C. A current of 150 mA was passed through a suspension for 30 min and 2 *M* HCl (2 mL) was added within the same time interval. The presence of water and ethanol played the decisive role. When the reaction was carried out in an acidic alcohol solution, only the reduction of P_4 to PH_3 was observed according to the earlier published report[29] and patent.[47]

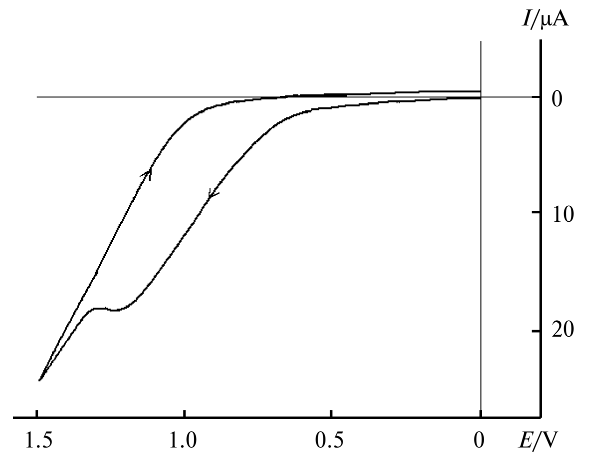
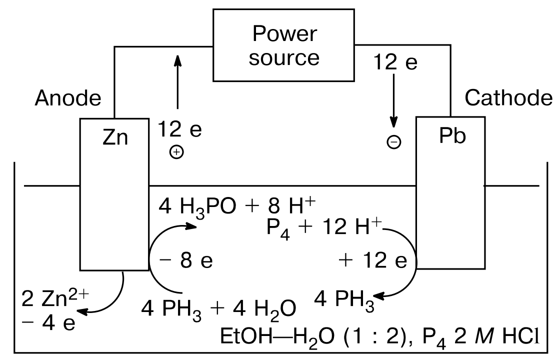


Fig. 1. Cyclic voltammogram of a saturated solution of PH_3 in ethanol (GC electrode, 0.1 *M* $(NBu_4)BF_4$).



Cathode

$$P_4 + 12\,H^+ + 12\,e \longrightarrow 4\,PH_3$$

Anode

$$\begin{cases} 4\,PH_3 + 4\,H_2O \longrightarrow 4\,H_3PO + 8\,H^+ + 8\,e \\ 2\,Zn \longrightarrow 2\,Zn^{2+} + 4\,e \end{cases}$$

$$\boxed{P_4 + 4\,H_2O + 2\,Zn + 4\,H^+ \longrightarrow 4\,H_3PO + 2\,Zn^{2+}}$$

Fig. 2. Electrochemical synthesis of H_3PO from P_4 in an H_2O—EtOH acidic solution.

The optimized process results in the full conversion of P_4 with the formation in the reaction mixture of three major products identified by signals in the $^{31}P\{^1H\}$ NMR spectra (Fig. 3, *a*). The signals with chemical shifts

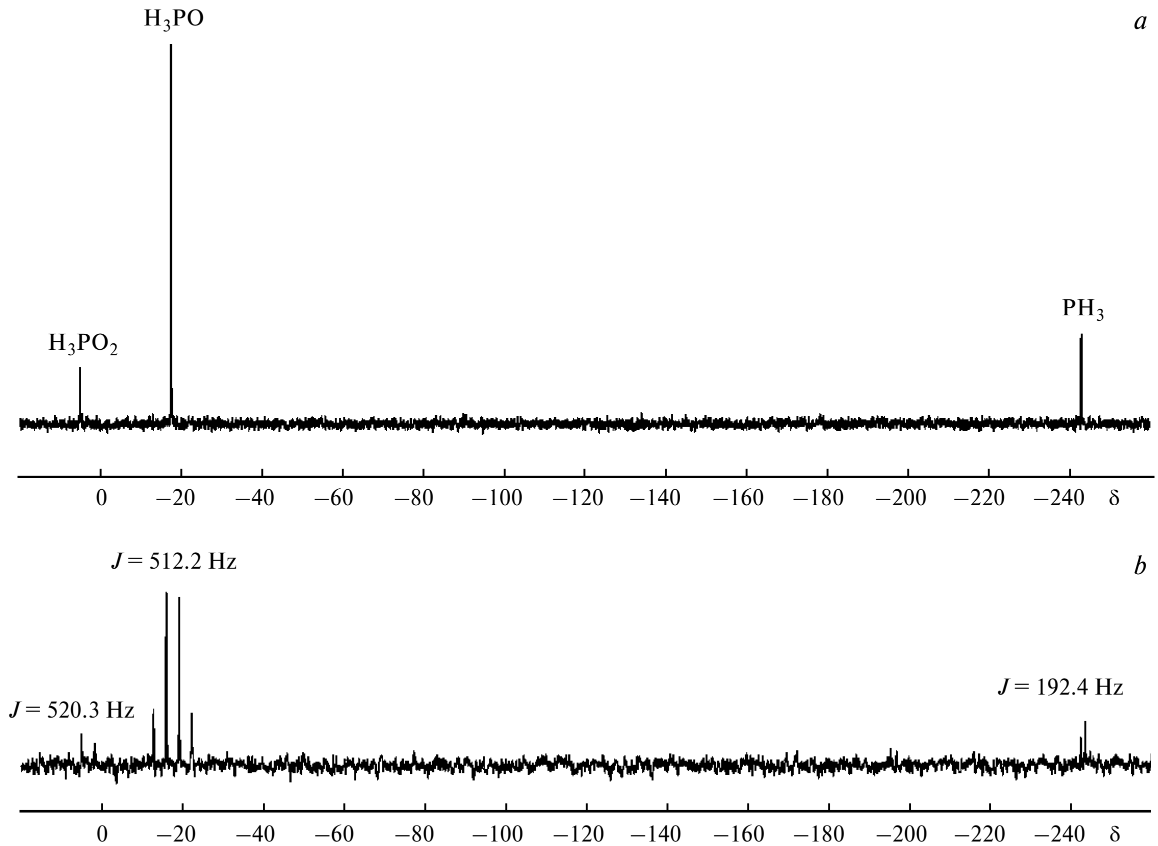


Fig. 3. $^{31}P\{^1H\}$ (*a*) and $^{31}P\{-\}$ (*b*) NMR spectra of the reaction mixture obtained by the electrochemical reduction of white phosphorus.

δ −243.05 and 4.98 were assigned to phosphine PH_3 (15.1%) and H_3PO_2 (16.1%), respectively, whereas the singlet at δ −17.56 was attributed to phosphine oxide H_3PO (68.8%). The registration of the ^{31}P NMR spectra without proton decoupling (Fig. 3, *b*) transforms the latter singlet into a quartet, as it was expected for three protons directly bound to the phosphorus atom ($^1J_{P-H}$ = 512.2 Hz).

In addition, the distinct cross-peak with the doublet having the center at δ 7.00 ($^1J_{P-H}$ = 512.2 Hz) in the 2D $^{31}P\{^1H\}$ NMR and $^1H-^{31}P$ HMQC NMR spectra unambiguously indicates the direct bound between the phosphorus and hydrogen nuclei leading to the appearance of ^{31}P resonance ascribed to H_3PO.

It should be mentioned that the disproportionation of electrochemically generated H_3PO to phosphine PH_3 and hypophosphorous acid H_3PO_2 prevents obtaining a more complete array of physicochemical data for phosphine oxide. However, it was established that the IR spectrum of the reaction mixture exhibits a new band at 1250 cm^{-1}, which was assigned to vibrations of the P=O group of phosphine oxide. Unfortunately, we were not able to get mass spectrometric data because of the fast decomposition of H_3PO under the experimental conditions.

Thus, it was found that electrochemically generated H_3PO slowly disproportionates in solution to form PH_3 and H_3PO_2 (Scheme 4). At −30 °C the process is slow, and after 20 days at −20 °C the H_3PO concentration is about a half of the initial one. At ambient temperature, the transformation is much faster ($T_{1/2} \approx$ 345 min) and H_3PO completely disproportionates to PH_3 and H_3PO_2 within approximately two days.

Scheme 4

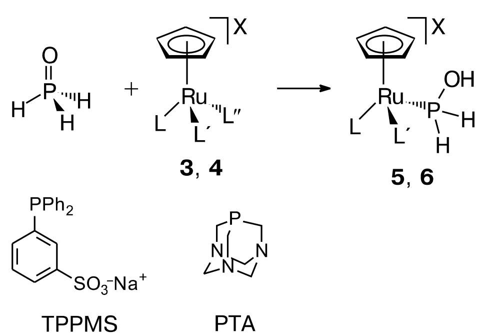
$$2\ H_3PO \longrightarrow PH_3 + H_3PO_2$$

Attempts to isolate phosphine oxide in the pure form from an aqueous ethanol solution were unsuccessful. After all volatiles were removed, *i.e.*, the solvent and PH_3, an oily residue was obtained and analyzed by $^{31}P\{^1H\}$ NMR spectroscopy. The presence of H_3PO along with a variable amount of H_3PO_2 was observed.

The structure of new compound **1** and the process of its electrochemical generation were confirmed by capturing this molecule as a thermally stable derivative.[46] According to the literature search concerning the reactivity of white phosphorus involving mediators of metals[11–13] showed that, after removal of chlorides from the coordination sphere, the ruthenium(II) complex [CpRu(PPh$_3$)$_2$Cl] (**2**) is capable of coordinating both white phosphorus itself and phosphine[48] and stabilizing the hydrolysis products of coordinated P_4,[49—51] including phosphorus oxyacids H_3PO_2 and H_3PO_3.[52] The presented phosphorus oxy derivatives can act as ligands, being transformed (due to ruthenium-promoted isomerization) into phosphine-like tautomers containing the three-coordinated phosphorus atom $HP(OH)_2$ and $P(OH)_3$, respectively.[52]

Thus, we suggested that water-soluble complex **2** can react with electrochemically generated H_3PO by the stabilization of assumed phosphonous acid $H_2P(OH)$ due to energetically favored H_3PO tautomerism at ruthenium. Indeed, the reaction of H_3PO with the ruthenium complexes $[CpRu(TPPMS)_2Cl]$ (**3**) (Cp is the cyclopentadienyl anion ($C_5H_5^-$), TPPMS is sodium triphenylphosphine monosulfonate *m*-$SO_3C_6H_4PPh_2^-Na^+$) (see Ref. 53) and $[CpRu(PTA)(MeCN)_2]PF_6$ (**4**) (PTA is 1,3,5-triaza-7-phosphaadamantane $P(CH_2)_3(NCH_2)_3$) (see Ref. 54) followed by the treatment of the reaction mixture afforded the corresponding organometallic complexes $[CpRu(TPPMS)_2\{H_2P(OH)\}]PF_6$ (**5**) and $[CpRu(PTA)(MeCN)\{H_2P(OH)\}]PF_6$ (**6**) (Scheme 5).

Scheme 5



Reagents and conditions: H_2O—EtOH (2 : 1), $TlPF_6$, 50 °C (for **5**) or 20 °C (for **6**).

Compound	L	L´	L´´	X
3	TPPMS	TPPMS	Cl	—
5	TPPMS	TPPMS	—	PF_6^-
4	PTA	MeCN	MeCN	PF_6^-
6	PTA	MeCN	—	PF_6^-

Since the reaction mixture in all cases contained hypophosphorous acid and phosphine, which are the disproportionation products of the H_3PO molecule, the formation of complexes $[CpRu(TPPMS)_2(PH_3)]PF_6$, $[CpRu(TPPMS)_2\{HP(OH)_2\}]PF_6$, $[CpRu(PTA)(MeCN)(PH_3)]PF_6$, and $[CpRu(PTA)(MeCN)\{HP(OH)_2\}]PF_6$ was observed in the reaction mixture. These complexes were identified in the reaction mixture using the data of NMR spectroscopy for the compounds synthesized independently by the reactions of complexes **3** and **4** with PH_3 and H_3PO_2, respectively.[46] Unfortunately, the presence of admixtures of these complexes did not allow one to obtain crystalline samples of complexes **5** and **6** suitable for X-ray diffraction analysis.

In conclusion, the elusive and poorly investigated phosphine oxide H_3PO (**1**) was generated in solution by simple electrochemical methods based on the two-stage process including the electrochemical reduction of white phosphorus to PH_3 in an acidic aqueous ethanol solution on the lead electrode and the electrochemical oxidation of PH_3 on the zinc anode. The optimized process makes it possible to obtain H_3PO with the maximum yield about 70%. The spontaneous disproportionation of **1** to PH_3 and hypophosphorous acid H_3PO_2 occurs even at low temperature and impedes the formation of H_3PO in the pure state. Nevertheless, the existence of the new, earlier undescribed phosphorus compound, *viz*., phosphine oxide H_3PO, in the free state was experimentally proved for the first time by NMR spectroscopy. This intermediate was isolated as a ligand in the coordination sphere of ruthenium(II) by tautomerism to phosphonous acid $H_2P(OH)$, giving the stable organometallic cationic ruthenium(II) complexes.

2. Electrochemical synthesis of organophosphorus compounds from white phosphorus

2.1. Electrochemical synthesis of OPCs from white phosphorus and organic substrates

The electrochemical formation of different types of organophosphorus compounds with P—O, P—N, and P—C bonds is reviewed in rather detail.[55] In this case, we will focus our attention on the most significant results obtained in the recent years when developing the methods for the preparation of OPCs with P—C bonds.

The presence of P—H groups in phosphine and intermediate phosphorus hydrides electrochemically generated from white phosphorus allowed their use for the preparative synthesis of OPCs from the latter. It was found that phosphine and a series of other P—H derivatives can react with various organic substrates,[35—39] including *in situ* interaction in the case of preparative electrolysis. A necessary condition for the target process of OPCs formation with P—C bonds is electrochemical inertness of the used olefins[56] and carbonyl compounds.[57,58] These reactions give mixtures of phosphines with an overall yield of 55—70% based on phosphorus or phosphine oxides in a 20—40% yield.

It is known that organic halides (RX), which are known as the main substrates of electrochemical coupling processes, are among the most convenient reagents for the introduction of the organic fragment to the phosphorus atom under the electrochemical conditions.[14] The combined reduction of white phosphorus with lower alkyl halides[59] affords quaternary phosphonium salts (29—40% yield). However, these compounds were isolated from the reaction mixture in the form of the corresponding phosphine oxides. It should be mentioned that the electrochemical reaction should include the electrochemical reduction of white phosphorus for the target process to occur. In the case of organyl halides with the reduction potential close to that of P_4, the process affords a series of organophosphorus products. It was shown[60] that the electrolysis of an emulsion of P_4 in DMF in the presence of butyl bromide (BuBr) affords a complicated mixture of mono-, di-, and tetraphosphines and phosphine oxides with an overall yield of 29% based on phosphorus. According to the mechanism proposed by authors, phosphorus, BuBr, and hydrogen are simultaneously reduced at the cathode. In addition, it was shown that the solvent can participate in the formation of organophosphorus products. However, the use of proton-donor media does not allow one to solve completely the problems concerning the industrially applicable production of OPCs from white phosphorus under the conditions of chlorine-free electrochemical synthesis. In all cases described above, a considerable portion of phosphorus is alkylated only partially (20—50%), but phosphorus is mainly polymerized. The low conversion of P_4 to OPCs and very complicated mixtures formed impede the isolation and purification of the products.

The results of practical and theoretical significance in the field of electrochemistry of white phosphorus were obtained at the Kazan Chemical School under the supervision of Prof. Yu. M. Kargin. It was experimentally established that the simultaneous treatment of an emulsion of white phosphorus with the electrochemically generated nucleophilic (ArO^-) and electrophilic (I_2) agents results in the conversion of white phosphorus and formation of aromatic esters of phosphorus acids.[61] It was shown in the same work that the functionalization of white phosphorus at the stage of transformation of the group with the P—H bond involves alcohol molecules to form normal or partial phosphorus acid esters. The method for electrochemical synthesis of trialkyl phosphates,[22] triamidophosphates, and triamidothiophosphates[62] from white phosphorus was developed and the mechanisms of the most probable reaction routes were proposed. The modification of conditions of the above presented experiments allowed one to enhance, in some cases, both the yields of target products and selectivity of the electrochemical reactions.[63—65] This combination of electrochemical reactions and reactions in a homogeneous medium made it possible to determine advantages of electrochemical synthesis methods over standard classical methods, which are accompanied, as a rule, by the polymerization of white phosphorus.[66]

Other industrial methods for OPCs synthesis from white phosphorus includes the use of undivided electrolyzers equipped with soluble metallic anodes. The possibility of undivided synthesis of OPCs from white phosphorus was demonstrated in the American patents.[67,68] In this case, the process occurs in solutions in alcohol, phenol, or mercaptane against the background of nitrates, bicarbonates, and other salts using either the mercury,[67] or ferrophosphorus anode.[68] Hydrogen is evolved on the cathode, and the metal cation formed upon the dissolution of the anode is an oxidant. However, the current efficiencies of triorganyl phosphates are low (45—50%), and the method is not "green." The main restriction of the method considered is the use of large amounts of mercury. Unfortunately, there is no data on the mechanism of formation of organophosphorus products.

2.2. Electrochemical synthesis of OPCs from white phosphorus in the presence of a mediator

To increase the selectivity of phosphorus—carbon bond formation under the electrochemical conditions from white phosphorus, it was proposed[23] to use a mediator system based on the nickel complexes stabilized by imine ligands of the 2,2´-bipyridine (bpy) type, which well recommended itself in various coupling processes involving organic halides and chlorophosphines.[14,69] Thus, the available theoretical concepts about the mechanisms of electrocatalytic coupling processes involving organic halides catalyzed by the nickel complexes with 2,2´-bipyridine were extended over other types of processes involving substrates containing ordinary bonds, one of which is white phosphorus.[7,69] The electrolysis of a solution containing organic halide and white phosphorus in the presence of the electrochemically generated catalysts, *viz.*, the nickel(0) complexes with 2,2´-bipyridine, allowed one to convert white phosphorus to compounds with P—C bonds, namely, phosphines and phosphine oxides, in rather high yields[23,69] (Table 1).

To enhance the selectivity of the formation of OPCs containing the tricoordinate phosphorus atom, it was proposed to use soluble metallic anodes of magnesium, aluminum, and zinc, which currently are well recommended in electrochemical processes, including electrocatalytic functionalization of organic halides (RX).[70] It should be mentioned that anodically generated metal ions can act as electrophilic components of the system favoring the stabilization of phosphide anions formed under the electroreduction conditions. The performed experiments showed that organic compounds of tricoordinate phosphorus containing three P—C bonds are predominantly formed under these conditions (see Table 1).

The active form of the catalyst (nickel(0) complex) is generated at the cathode during the electrochemical process, and the oxidation of the material of the soluble anode is used in the anodic reaction (Scheme 6).

Similarly to the earlier studied cases of P_4 functionalization by Grignard reagents,[71] it can be assumed that the

Table 1. Organophosphorus products of electrolysis of a solution of white phosphorus and organic halides (RX) in DMF in the presence of the nickel complexes with 2,2´-bipyridine ligand

R—X	Anode	Products	Yield* by P_4 (%)
PhI	GC (membrane)	Ph_3P	11
		Ph_3PO	75
	Zn	Ph_3P	65
		Ph_2PH	12
		$PhPH_2$	10
	Mg	Ph_3P	50
		Ph_3PO	10
	Al	Ph_3P	40
		Ph_3PO	18
PhBr	GC (membrane)	Ph_3P	10
		Ph_3PO	79
	Zn	Ph_3P	50
		Ph_3PO	10
		Ph_2PH	9
	Mg	$(PhP)_5$	60
		Ph_3P	15
	Al	Ph_3P	38
		Ph_3PO	10

* The yields of primary and secondary phosphines were determined on the basis of the calibration plots of the integral intensities of signals in the $^{31}P\{-\}$ NMR spectra after the extraction of organophosphorus products.

Scheme 6

Cathode

$$[Ni^{2+}(bpy)_2] + 2\,e^- \xrightarrow{E_p{}^{C_1}} [Ni^0(bpy)_2]$$

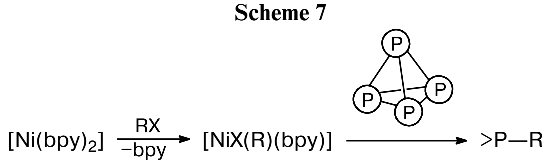
Anode

$$M^0 \longrightarrow M^{n+} + ne$$

n = 2 (Ni, Mg, Zn), 3 (Al)

key stage of P—C bond formation is the interaction of the organonickel σ-complex [NiX(Ar)(bpy)] generated electrochemically in the reaction medium with the white phosphorus molecule (Scheme 7).

Scheme 7



It was found that the full or partial recovery of the nickel(II) complex with 2,2´-bipyridine is observed during preparative electrolysis, depending on the nature of RX and experimental conditions.

In order to confirm this reaction mechanism (see Scheme 7), we electrochemically synthesized[72] stable organonickel σ-complex of the [NiBr(Ar)(bpy)] type[7—9] (Ar is 2,4,6-trimethylphenyl (**7**), 2,4,6-triisopropylphenyl (**8**), and 2,6-dimethylphenyl (**9**)), which were introduced in the reaction with white phosphorus. However, it was found that the organonickel σ-complexes are absolutely inert with respect to the P_4 molecule. Mixing of solutions of white phosphorus and the organonickel σ-complex in diverse solvents (DMF, benzene, toluene, and THF) and heating of the reaction mixtures to 60 °C did not result in any interaction. It was established that the reaction occurs only upon the electrochemical activation of the organonickel σ-complex molecule due to the one-electron reduction (Table 2).

Previously A. Klein showed[73] for the organonickel σ-complex [NiBr(Mes)(bpy)] that the electron transfer to the molecule is accompanied by the elimination of the bromide anion with the formation of the neutral coordinatively unsaturated radical complex capable of coordinating the substrate molecule (Scheme 8).

Scheme 8

$$[NiBr(Mes)(bpy)] + e \longrightarrow [Ni(Mes)(bpy)]^{\cdot} + Br^-$$

This interaction results in the formation of the active form of the complex due to the elimination of the bromide anion, initiating the coordination of the P_4 molecule to the nickel center followed by reductive elimination with P—C bond formation (Scheme 9).

Scheme 9

$$[Ni^{2+}(bpy)] + 2\,e \rightleftharpoons [Ni^0(bpy)]$$

$$[Ni^0(bpy)] + ArBr \longrightarrow [NiBr(Ar)(bpy)]$$

$$[NiBr(Ar)(bpy)] + e \rightleftharpoons [Ni(Ar)(bpy)] + Br^-$$

$$[Ni(Ar)(bpy)] + P_4 \longrightarrow [Ni(Ar)(P_4)(bpy)] \longrightarrow$$

$$\longrightarrow [Ar—P<] + [Ni^0(bpy)]$$

Insoluble organic nickel phosphides are simultaneously formed and can readily be transferred to the solution upon decomposition with mineral acids with the oxidation of the phosphorus center (Scheme 10). Thus, in the case of σ-complexes **7** and **8**, arylphosphonous and arylphosphinic acids were obtained, which confirms formation of com-

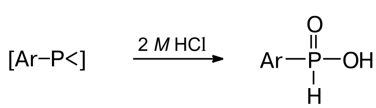
Table 2. Potentials* of peaks (E_p) and currents (I) on the CV curves of organonickel σ-complexes **7** and **8** observed at the scanning of the working electrode (GC) potential to the cathodic region

Complex	Cathodic peaks			Anodic peaks (re-oxidation)		
	Peak	$-E_p$/V	I/μA	Peak	$-E_p$/V	I/μA
[NiBr(Mes)(bpy)] (**7**)	C_1	1.80	(4.8)	A_1	1.64	(1.2)
	C_2	2.15	(4.7)	A_2	2.05	(4.6)
	C_3	2.62	(0.4)	A_3	2.50	(0.4)
[NiBr(Tipp)(bpy)] (**8**)	C_1	1.75	(5.9)	A_1	1.54	(0.6)
	C_2	2.03	(2.0)	A_2	1.97	(4.2)
	C_3	2.33	(1.3)	A_3	2.20	(1.0)

* The CV curves were detected without IR compensation, and the potentials are presented with respect to $Ag/AgNO_3$, 0.01 *M* in MeCN system.

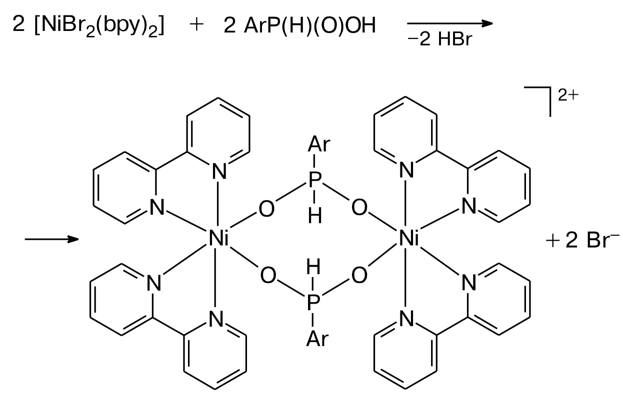
pounds with P—C bonds under the electrochemical conditions *via* the reaction of the organonickel σ-complexes with white phosphorus.

Scheme 10



It was experimentally established that arylphosphonous acids ArP(H)(O)OH can be also formed under the electrolysis conditions in the presence of trace amounts of water and oxygen. The compounds formed can deactivate the active form of the organonickel catalyst $[Ni^{2+}(bpy)_2]$ due to the formation of bimetallic nickel complexes **10** and **11** insoluble in organic solvents (Scheme 11). Their structures were determined by X-ray diffraction analysis (Fig. 4).[74]

Scheme 11



10, **11**

Ar = Ph (**10**), Mes (**11**)

Thus, the reaction of the electrochemically activated organonickel σ-complexes with white phosphorus affords

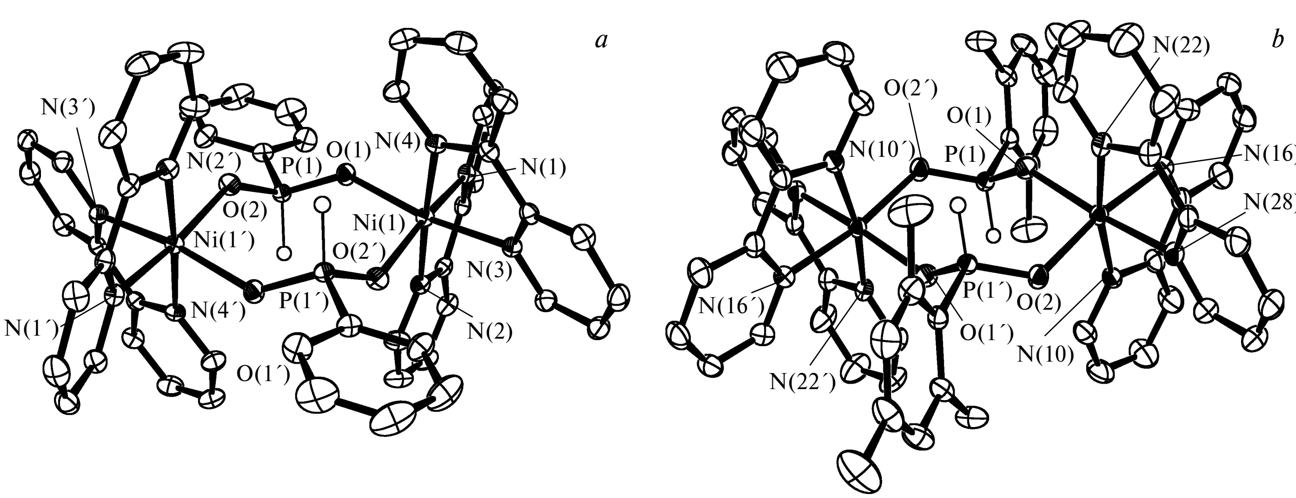


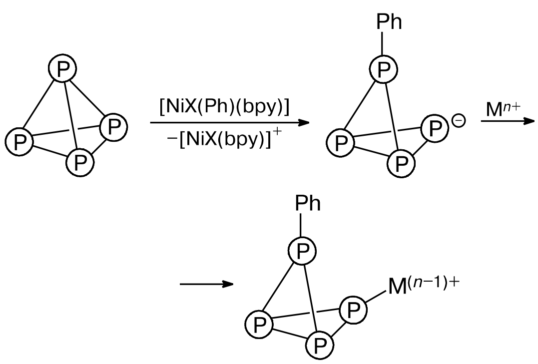
Fig. 4. Crystal structures of bimetallic cations $[Ni_2(\mu\text{-}O_2P(H)Ph)_2(bpy)_4]^{2+}$ (**10**) (*a*) and $[Ni_2(\mu\text{-}O_2P(H)Mes)_2(bpy)_4]^{2+}$ (**11**) (*b*).

organophosphorus compounds with P—C bonds. However, the cyclic recovery of the catalyst and the presence of electrophilic components in the system (organyl halides, anodically generated metal ions), which provide the stabilization of phosphide anions formed at the stage of P—P bond cleavage in the white phosphorus molecule, are necessary for the successful occurrence of the electrochemical process and formation of OPCs containing two or three phosphorus—carbon bonds.

The results of preparative syntheses showed that the material of the soluble anode substantially affected the nature of electrolysis products (Fig. 5).[75]

The anodically generated metal cations were found to act as electrophilic agents of the system, since they stabilize phosphide anions formed as a result of the primary attack of the organonickel σ-complexes electrochemically generated in the reaction medium to a white phosphorus molecule (Scheme 12).

Scheme 12



Thus, the side processes of phosphorus polymerization and formation of insoluble precipitates of polyphosphides can be avoided. The experimental data indicate an important role of metal ions in the mechanism of phosphorus

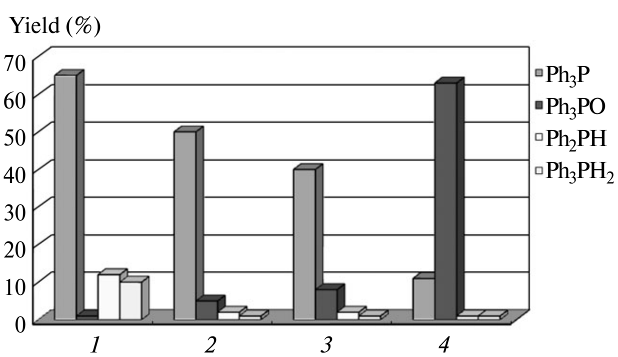


Fig. 5. Yields of the organophosphorus products obtained by the preparative electrochemical reduction of white phosphorus in the presence of iodobenzene and complex $[Ni(bpy)_3](BF_4)_2$ in DMF on the platinum electrode (cathode) using Zn (*1*), Mg (*2*), and Al (*3*) anodes and in a divided electrolyzer (*4*).

tetrahedron opening and transformation of phosphorus oligomers into the final products, and the purposed choice of the material of the soluble anode allows one to control, to a certain extent, the nature and yield of the final products of electrochemical arylation of P_4. Especially attractive results were obtained with the zinc anode and, hence, this process was studied in more detail.

2.3. Electrochemical synthesis of triphenylphosphine involving zinc(II) ions

The use of the Zn anode leads to the complete conversion of P_4 to soluble phosphorus compounds (mainly tertiary phosphines R_3P) and, to a lesser extent, to mono- and diarylphosphines. Among the known metal phosphides, only phosphides of zinc and alkaline metals efficiently react with RX (see Ref. 76) to give OPCs with P—C bonds. Nickel, aluminum, and magnesium phosphides are considerably less reactive. Therefore, the contribution of nickel phosphide formation to the P_4 tetrahedron opening and its subsequent functionalization with the formation of OPCs bearing P—C bonds is rather low.

Another possibility of participation of Zn^{2+} ions generated on the anode in the reactions of catalytic transformation of P_4 is transmetallation: the substitution of nickel in the organonickel σ-complexes.[77] The cross-coupling reactions of RX including the stage of catalytic formation of organozinc compounds were described[78] (Scheme 13).

Scheme 13

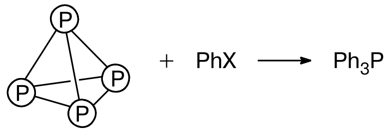
$$[NiBr(R)(bpy)] + Zn^{2+} \longrightarrow [ZnBr(R)(bpy)] + Ni^{2+}$$

However, there is no data on the occurrence of this reaction. Nevertheless, the fact of formation of organozinc compounds from RX and zinc salts under the catalytic conditions in the presence of the Ni^0 complex is doubtless, since complex [ZnX(R)(bpy)] was used, in many cases, for the functionalization of various substrates.[78—80]

It was found that the nature of the soluble anode not only substantially affects the nature and yield of the final products of electrochemical phosphorylation of organic halides by white phosphorus but also results in the formation of organophosphorus compounds in the absence of an organonickel catalyst during electrolysis in an electrolyzer with undivided anodic and cathodic spaces and equipped with the electrochemically soluble zinc anode.[81] Under these conditions, the electrolysis of a solution containing white phosphorus and aromatic halide (PhI, PhBr) results in the selective formation of triphenylphosphine in the yields of the target product up to 82% (Scheme 14).

In order to study the mechanism and main intermediates of the electrochemical synthesis of triphenylphosphine from organic halides and white phosphorus in an

Scheme 14



X = I, Br

Conditions: Pt cathode, Zn anode, DMF, $E = E(Zn^{2+}/Zn^0) = -1.60—-1.90$ V (*vs* Ag/AgNO$_3$, 0.01 *M* in MeCN), $I = 8$ mA cm^{-2}.

undivided electrolyzer equipped with the soluble zinc anode, we examined the electrochemical properties of anodically generated zinc(II) ions both in the absence and in the presence of organic halides and white phosphorus.

An analysis of the voltammetric curves of the systems containing zinc bromide, supporting electrolyte, and solvent and the preparative electrolysis data show that the anodically generated zinc(II) ions are capable of electrochemical reducing to form metallic zinc adsorbed on the operating electrode surface. However, in the presence of organic halides, electrochemical reduction affords organozinc compounds, which also reflects the CV curves of the systems: the anodic peak of oxidation of adsorbed metallic zinc disappears.[82] It should be mentioned that this mechanism of formation of organometallic compounds is also characteristic of the mercury-based systems.[83] The disappearance of the anodic peaks of metallic zinc oxidation can be a consequence of the fast interaction of electrochemically generated zinc(0) with organic halide already in the near-electrode layer to form the organozinc product. The accumulation of organozinc derivatives near the working electrode surface creates, in turn, diffusional hindrances for the electrochemical reduction of the initial zinc(II) ions that migrate to the surface of the working electrode. In this case, one should take into account the possibility of electrode passivation, as in the case of the earlier described processes on the Pt and Zn electrodes[84] for which radical processes of reduction of Zn^{2+} ions proceeding through the stage of radical ion formation followed by solvent polymerization results apparently in the blocking of the working electrode surface.

Thus, based on the electrochemical studies performed and obtained experimental data it was concluded that the electrochemical reduction of zinc(II) ions affords Zn^0 species capable of entering oxidative addition reactions with organic halides to form organozinc compounds. The rate of this process depends on both the nature of organyl halide used and the nature of the solvent used.

The addition of white phosphorus to a solution containing Zn^{2+} ions does not change the CV curve. Therefore, it can be concluded that the reaction between electrochemically generated zinc(0) and white phosphorus does not occur with a sufficient rate.

The synthesized organozinc compounds [Ph_2Zn] and [PhZnBr] were used for the investigation of the reactivity of organozinc compounds towards white phosphorus.[85] The interaction of white phosphorus with these organozinc derivatives was established to lead to the formation of organic phosphides that, being decomposed with water, give primary and secondary phosphines $PhPH_2$ and Ph_2PH. Similar attempts to arylate and alkylate elemental (white) phosphorus by the action of organometallic agents were made earlier for the magnesium and lithium derivatives.[71] Rather high yield of phenylphosphine (25—40%) was attained in the reaction of white phosphorus with phenylmagnesium bromide or phenyllithium.[71]

Thus, the obtained experimental data on the electrochemical properties of the anodically generated zinc(II) ions and reactivity of the organozinc reagents with respect to white phosphorus suggest the possible scheme of electrochemical synthesis of triphenylphosphine from white phosphorus and organic halides in an undivided electrochemical cell equipped with the electrochemically soluble zinc anode. At the first stage of the process, the anodically generated Zn^{2+} ions are electrochemically reduced to form Zn(0) species, which react with the organic halide present in the solution to give organozinc compounds capable of reacting with the P_4 molecule to form the organophos-

Scheme 15

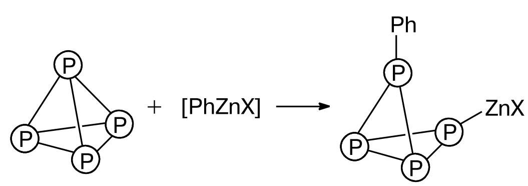
Electrochemical generation of Zn(0)

$$Zn^{2+} + 2\,e \longrightarrow Zn^0, \qquad (1)$$

Formation of organozinc compounds

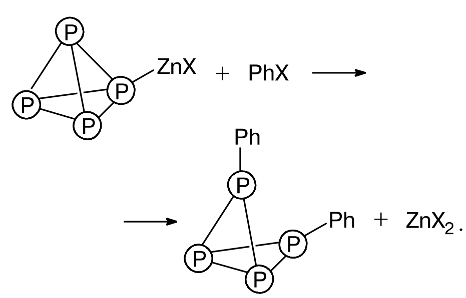
$$Zn^0 + PhX \longrightarrow [PhZnX], \qquad (2)$$

Cleavage of P—P bonds in P_4 molecule and formation of P—C bonds



(3)

Interaction of zinc phosphides with organic halides



(4)

phorus derivatives bearing P—C bonds and zinc phosphides. It is known[76] that of all metal phosphides only zinc and alkaline metal phosphides can react with organic halides to form compounds bearing P—C bonds. Thus, the reaction of the formed organic zinc phosphides with organic halides affords new phosphorus—carbon bonds (Scheme 15).

The further consecutive interaction during the electrochemical process results in the cleavage of all P—P bonds in the white phosphorus molecule to form organophosphorus compounds containing three phosphorus—carbon bonds at the phosphorus atom.

Thus, elemental (white) phosphorus is an electrochemically active compound and easily enters electrochemical oxidation/reduction reactions both in the absence and in the presence of organic compounds. The nature of the formed products of electrochemical transformations strongly depends on the electrolysis conditions. The change of the conditions (solvent, electrode material, temperature, and pH of the solution) results in hitherto unknown, highly reactive organophosphorus compounds from white phosphorus.

References

1. *Technology Vision 2020. The U.S. Chemical Industry*, Am. Chem. Soc., Washington, 1996, 75 pp.
2. *Studies in Inorganic Chemistry 20. Phosphorus. An outline of its Chemistry, Biochemistry and Technology*, 5 ed., Ed. D. E. C. Corbridge, Elsevier (Science B. V.), Amsterdam—Lausanne—New York—Oxford—Shannon—Tokyo, 1995, 1208 pp.
3. F. Mathey, *Angew. Chem.*, *Int. Ed.*, 2003, **42**, 1578.
4. F. Mathey, *Angew. Chem.*, *Int. Ed.*, 1987, **26**, 275.
5. V. A. Milyukov, Yu. H. Budnikova, O. G. Sinyashin, *Russ. Chem. Rev.*, 2005, **74**, 781 [*Usp. Khim.*, 2005, **74**, 859].
6. M. Peruzzini, L. Gonsalvi, A. Romerosa, *Chem. Soc. Rev.*, 2005, **34**, 1038.
7. Yu. H. Budnikova, D. G. Yakhvarov, O. G. Sinyashin, *J. Organomet. Chem.*, 2005, **690**, 2416.
8. G. P. Chiusoli, P. M. Maitlis, *Metal-catalysis in Industrial Organic Processes*, Royal Society of Chemistry, Cambridge, 2006, p. 290.
9. I. P. Beletskaya, L. M. Kustov, *Russ. Chem. Rev.*, 2010, **79**, 441 [*Usp. Khim.*, 2010, **79**, 493].
10. H. Lund, *J. Electrochem. Soc.*, 2002, **149**, S21.
11. M. Caporali, L. Gonsalvi, A. Rossin, M. Peruzzini, *Chem. Rev.*, 2010, **110**, 4178.
12. B. M. Cossairt, N. A. Piro, C. C. Cummins, *Chem. Rev.*, 2010, **110**, 4164.
13. M. Scheer, G. Balazs, A. Seitz, *Chem. Rev.*, 2010, **110**, 4236.
14. Yu. H. Budnikova, *Russ. Chem. Rev.*, 2002, **71**, 111 [*Usp. Khim.*, 2002, **71**, 126].
15. J.-Y. Nedelec, J. Perichon, M. Troupel, *Organic Electroreductive Coupling Reactions using Transition Metal Complexes as Catalysts*, in *Topics in Current Chemistry*, Ed. E. Steckhan, Springer-Verlag, Berlin—Heidelberg, 1997, **185**, p. 141.
16. T. V. Magdesieva, *Elektrokhimiya metalloorganicheskikh soedinenii i metallokompleksov* [*Electrochemistry of Organometallic Compounds and Metal Complexes*], in *Elektrokhimiya organicheskikh soedinenii v nachale XXI veka* [*Electrochemistry of Organic Compounds in the Beginning of the XXI Century*], Eds V. P. Gul´tyai, A. G. Krivenko, A. P. Tomilov, Kompaniya Sputnik+, Moscow, 2008, 250 pp. (in Russian).
17. Y. Tamaru, *Modern Organonickel Chemistry*, Wiley-VCH, Weinheim, 2005, 346.
18. J. Campora, *Nickel-carbon σ-Bonded Complexes*, in *Comprehensive Organometallic Chemistry III*, Elsevier, Amsterdam—Boston, 2007, **8**, 27.
19. P. W. Jolly, G. Wilke, *The Organic Chemistry of Nickel*, Acad. Press, London, 1975, Vol. **2**, p. 94.
20. A. P. Tomilov, *Moi put´ v nauke* [*My Way in Science*], Khoruzhevskii A. I., Moscow, 2009, 184 pp. (in Russian).
21. I. N. Brago, A. P. Tomilov, *Elektrokhimiya*, 1968, **4**, 697 [*Sov. Electrochem.* (*Engl. Transl.*), 1968, **4**, No. 6].
22. A. S. Romakhin, Yu. G. Budnikova, I. M. Zaripov, Yu. M. Kargin, E. V. Nikitin, A. P. Tomilov, Yu. A. Ignat´ev, V. V. Smirnov, *Bull. Russ. Acad. Sci., Div. Chem. Sci.* (*Engl. Transl.*), 1992, **41**, 1031 [*Izv. Akad. Nauk*, *Ser. Khim.*, 1992, 1322].
23. Yu. G. Budnikova, D. G. Yakhvarov, Yu. M. Kargin, *Mendeleev Commun.*, 1997, **7**, 67.
24. M. K. Kadirov, Yu. H. Budnikova, K. V. Kholin, M. I. Valitov, S. A. Krasnov, T. V. Gryaznova, O. G. Sinyashin, *Russ. Chem. Bull.* (*Int. Ed.*), 2010, **59**, 466 [*Izv. Akad. Nauk*, *Ser. Khim.*, 2010, 456].
25. V. V. Turygin, A. V. Khudenko, A. P. Tomilov, *Russ. J. Electrochem.* (*Engl. Transl.*), 1999, **35**, 303 [*Elektrokhimiya*, 1999, **35**, 333].
26. I. M. Osadchenko, A. P. Tomilov, *Zh. Prikl. Khim.*, 1970, **43**, 1255 [*J. Appl. Chem. USSR* (*Engl. Transl.*), 1970, **43**, No. 6].
27. V. V. Turygin, A. P. Tomilov, M. Yu. Berezkin, V. A. Fedorov, *Inorg. Mater.*, 2010, **46**, 1459.
28. A. P. Tomilov, I. N. Brago, I. M. Osadchenko, *Elektrokhimiya*, 1968, **4**, 1153 [*Sov. Electrochemistry* (*Engl. Transl.*), 1968, **4**, No. 10].
29. N. Ya. Shandrinov, A. P. Tomilov, *Elektrokhimiya*, 1968, **4**, 237 [*Sov. Electrochemistry* (*Engl. Transl.*), 1968, **4**, No. 2].
30. I. M. Osadchenko, A. P. Tomilov, *Elektrokhimiya*, 1993, **29**, 406 [*Russ. J. Electrochem.* (*Engl. Transl.*), 1993, **29**, No. 3].
31. US Pat. 4021321; *Chem. Abstr.*, 1977, **87**, 13505.
32. US Pat. 4021322; *Chem. Abstr.*, 1977, **87**, 13505.
33. M. L. Barry, Ch. W. Tobias, *Electrochem. Technol.*, 1966, **4**, 502.
34. S. P. Varshavskii, A. P. Tomilov, Yu. D. Smirnov, *Zh. Vsesoyuz. Khim. Ob-va im. D. I. Mendeleeva*, 1962, **7**, 598 [*Mendeleev Chem. J.* (*Engl. Transl.*), 1962, **7**, No. 5].
35. B. A. Trofimov, S. N. Arbuzova, N. K. Gusarova, *Russ. Chem. Rev.*, 1999, **68**, 215 [*Usp. Khim.*, 1999, **68**, 240].
36. S. F. Malysheva, S. N. Arbuzova, in *Sovremennyi organicheskii sintez* [*Modern Organic Synthesis*], Ed. D. Rakhmankulov, Khimiya, Moscow, 2003, p. 160 (in Russian).
37. B. A. Trofimov, B. G. Sukhov, S. F. Malysheva, N. K. Gusarova, *Kataliz v promyshlennosti*, 2006, No. 4, 18 [*Catal. Ind.* (*Engl. Transl.*), 2006, No. 4].
38. B. A. Trofimov, N. K. Gusarova, *Mendeleev Commun.*, 2009, **19**, 295.

39. N. K. Gusarova, S. N. Arbuzova, B. A. Trofimov, *Pure Appl. Chem.*, 2012, **84**, 439.
40. L. R. Thorne, V. G. Anicich, S. S. Prasad, W. T. Huntress, *Chem. Soc. Rev.*, 2005, **34**, 691.
41. E. Maciá, *Chem. Soc. Rev.*, 2005, **34**, 691.
42. P. A. Hamilton, T. P. Murrells, *J. Chem. Soc.*, *Faraday Trans.*, 1985, **81**, 1531.
43. R. Withnall, L. Andrews, *J. Phys. Chem.*, 1987, **91**, 784.
44. R. Withnall, L. Andrews, *J. Phys. Chem.*, 1988, **92**, 4610.
45. D. A. Kayser, B. S. Ault, *J. Phys. Chem.*, 2003, **107**, 6500.
46. D. Yakhvarov, M. Caporali, L. Gonsalvi, Sh. Latypov, V. Mirabello, I. Rizvanov, O. Sinyashin, P. Stoppioni, M. Peruzzini, *Angew. Chem., Int. Ed.*, 2011, **50**, 5370.
47. Pat. Jpn. 01-139781; http://www19.ipdl.inpit.go.jp/PA1/cgi-bin/PA1DETAIL.
48. M. Di Vaira, P. Frediani, S. Seniori Costantini, M. Peruzzini, P. Stoppioni, *J. Chem. Soc., Dalton Trans.*, 2005, 2234.
49. P. Barbaro, M. Di Vaira, S. Seniori Costantini, M. Peruzzini, P. Stoppioni, *Angew. Chem., Int. Ed.*, 2008, **47**, 4425.
50. P. Barbaro, M. Di Vaira, S. Seniori Costantini, M. Peruzzini, P. Stoppioni, *Inorg. Chem.*, 2009, **48**, 1091.
51. M. Di Vaira, M. Peruzzini, P. Stoppioni, *C. R. Chim.*, 2010, **13**, 935.
52. D. N. Akbayeva, M. Di Vaira, S. Seniori Costantini, M. Peruzzini, P. Stoppioni, *J. Chem. Soc., Dalton Trans.*, 2006, 389.
53. A. Romerosa, M. Saoud, T. Campos-Malpartida, C. Lidrissi, M. Serrano-Ruiz, M. Peruzzini, J. A. Garrido, F.García-Maroto, *Eur. J. Inorg. Chem.*, 2007, 2803.
54. S. Bolaño, L. Gonsalvi, F. Zanobini, F. Vizza, V. Bertolasi, A. Romerosa, M. Peruzzini, *J. Mol. Catal. A: Chem.*, 2004, **224**, 61.
55. Yu. M. Kargin, Yu. H. Budnikova, B. I. Martynov, V. V. Turygin, A. P. Tomilov, *J. Electroanal. Chem.*, 2001, **507**, 157.
56. L. V. Kaabak, N. Ya. Shandrinov, A. P. Tomilov, *Zh. Obshch. Khim.*, 1970, **40**, 584 [*J. Gen. Chem. USSR* (*Engl. Transl.*), 1970, **40**, No. 3].
57. I. M. Osadchenko, A. P. Tomilov, *Zh. Obshch. Khim.*, 1969, **39**, 469 [*J. Gen. Chem. USSR* (*Engl. Transl.*), 1969, **39**, No. 2].
58. I. M. Osadchenko, A. P. Tomilov, *Zh. Obshch. Khim.*, 1970, **40**, 698 [*J. Gen. Chem. USSR* (*Engl. Transl.*), 1970, **40**, No. 3].
59. L. F. Filimonova, L. V. Kaabak, A. P. Tomilov, *Zh. Obshch. Khim.*, 1969, **39**, 2174 [*J. Gen. Chem. USSR* (*Engl. Transl.*), 1969, **39**, No. 10].
60. L. V. Kaabak, M. I. Kabachnik, A. P. Tomilov, S. L. Varshavskii, *Zh. Obshch. Khim.*, 1966, **36**, 2060 [*J. Gen. Chem. USSR* (*Engl. Transl.*), 1966, **36**, No. 12].
61. Yu. G. Budnikova, Yu. M. Kargin, I. M. Zaripov, A. S. Romakhin, Yu. A. Ignat´ev, E. V. Nikitin, A. P. Tomilov, *Bull. Russ. Acad. Sci.*, *Div. Chem. Sci.* (*Engl. Transl.*), 1992, **41**, 1585 [*Izv. Akad. Nauk*, *Ser. Khim.*, 1992, 2039].
62. Yu. G. Budnikova, Yu. M. Kargin, *Zh. Obshch. Khim.*, 1995, **65**, 1663 [*Russ. J. Gen. Chem.* (*Engl. Transl.*), 1995, **65**, No. 10].
63. Yu. G. Budnikova, Yu. M. Kargin, I. M. Zaripov, A. S. Romakhin, Yu. A. Ignat´ev, E. V. Nikitin, A. P. Tomilov, V. V. Smirnov, *Bull. Russ. Acad. Sci., Div. Chem. Sci.* (*Engl. Transl.*), 1992, **41**, 1580 [*Izv. Akad. Nauk*, *Ser. Khim.*, 1992, 2033].
64. Yu. G. Budnikova, Yu. M. Kargin, *Zh. Obshch. Khim.*, 1995, **65**, 566 [*Russ. J. Gen. Chem.* (*Engl. Transl.*), 1995, **65**, No. 4].
65. Yu. G. Budnikova, D. I. Tazeev, A. G. Kafiyatullina, D. G. Yakhvarov, V. I. Morozov, N. K. Gusarova, B. A. Trofimov, O. G. Sinyashin, *Russ. Chem. Bull.* (*Int. Ed.*), 2005, **54**, 942 [*Izv. Akad. Nauk*, *Ser. Khim.*, 2005, 919].
66. A. S. Romakhin, I. M. Zaripov, Yu. G. Budnikova, Yu. M. Kargin, E. V. Nikitin, A. P. Tomilov, Yu. A. Ignat´ev, *Bull. Russ. Acad. Sci.*, *Div. Chem. Sci.* (*Engl. Transl.*), 1992, **41**, 1036 [*Izv. Akad. Nauk*, *Ser. Khim.*, 1992, 1328].
67. Pat. US 4337125, 1982; http://ip.com/pat/US4337125.
68. Pat. US 4338166, 1982; http://ip.com/pat/US4338166.
69. D. G. Yakhvarov, Yu. H. Budnikova, O. G. Sinyashin, *Russ. J. Electrochem.* (*Engl. Transl.*), 2003, **39**, 1261 [*Elektrokhimiya*, 2003, **39**, 1407].
70. J. Chaussard, J. C. Folest, J. Y. Nedelec, J. Perichon, S. Sibille, *Synthesis*, 1990, 369.
71. M. M. Rauhut, A. M. Semsel, *J. Org. Chem.*, 1964, **28**, 471.
72. D. G. Yakhvarov, E. A. Trofimova, I. Kh. Rizvanov, O. S. Fomina, O. G. Sinyashin, *Russ. J. Electrochem.* (*Engl. Transl.*), 2011, **47**, 1100 [*Elektrokhimiya*, 2011, **47**, 1180].
73. A. Klein, A. Kaiser, B. Sarkar, M. Wanner, J. Fiedler, *Eur. J. Inorg. Chem.*, 2007, 965.
74. D. Yakhvarov, E. Trofimova, O. Sinyashin, O. Kataeva, Yu. Budnikova, P. Lönnecke, E. Hey-Hawkins, A. Petr, Yu. Krupskaya, V. Kataev, R. Klingeler, B. Büchner, *Inorg. Chem.*, 2011, **50**, 4553.
75. D. G. Yakhvarov, Yu. H. Budnikova, D. I. Tazeev, O. G. Sinyashin, *Russ. Chem. Bull.* (*Int. Ed.*), 2002, **51**, 2059 [*Izv. Akad. Nauk*, *Ser. Khim.*, 2002, 1903].
76. K. Issleib, *Z. Chem.*, 1962, **2**, 163.
77. Yu. H. Budnikova, D. I. Tazeev, D. G. Yakhvarov, *Russ. Chem. Bull.* (*Int. Ed.*), 2002, **51**, 269 [*Izv. Akad. Nauk*, *Ser. Khim.*, 2002, 256].
78. S. Durandetti, S. Sibille, J. Perichon, *J. Org. Chem.*, 1989, **54**, 2198.
79. S. Sibille, E. d´Incan, L. Leport, M. C. Massebiau, J. Perichon, *Tetrahedron Lett.*, 1987, **28**, 55.
80. A. Conan, S. Sibille, J. Perichon, *J. Org. Chem.*, 1991, **56**, 2018.
81. Pat. RF 2221805; www.fips.ru.
82. D. G. Yakhvarov, Yu. S. Ganushevich, A. B. Dobrynin, D. V. Krivolapov, I. A. Litvinov, O. G. Sinyashin, *Russ. J. Electrochem.* (*Engl. Transl.*), 2009, **45**, 139 [*Elektrokhimiya*, 2009, **45**, 148].
83. A. P. Tomilov, L. G. Feoktistov, *Elektrokhimiya organicheskikh soedinenii* [*Electrochemistry of Organic Compounds*], Mir, Moscow, 1976, p. 216 (in Russian).
84. S. Biallozor, E. T. Bandura, *Electrochim. Acta*, 1987, **32**, 891.
85. Pat. US 6258967; http://ip.com/pat/US6258967.

Received April 10, 2012;
in revised form July 2, 2012